

Trained endpoints w_A, w_B

3 architectures $\times$ 3 parameterisations $\times$ 4 activations $\times$ 7 widths $n \in \{64, \dots, 4096\} \times S \geq 3$ seed pairs
gauge-fixed post hoc by layerwise weight matching Π (never during training)

$$\Delta = \Pi w_B - w_A, \quad \gamma_{\text{lin}}(t) = w_A + t\Delta, \quad t \in [0, 1]$$

loss only:
never touches F

Matrix-free Fisher-vector product $v \mapsto Fv$

$$v \xrightarrow{\text{JVP}} u = J(x; w)v \xrightarrow{O(K)} a = p \odot u - p(p^\top u) \xrightarrow{\text{VJP}} J(x; w)^\top a = Fv$$

$O(P)$ memory: $J \in \mathbb{R}^{K \times P}$ and $F \in \mathbb{R}^{P \times P}$ are never formed

Barrier $B(\gamma_{\text{lin}})$

≥ 101 -point grid
+ local refinement,
grid-doubling check,
floor $\frac{1}{2}|L(w_A) - L(w_B)|$

Fisher length

$$Q_F = \Delta^\top F \Delta$$

unregularised F

Rayleigh quotient

$$\hat{\Delta}^\top F \hat{\Delta}$$

$$\hat{\Delta} = \Delta / \|\Delta\|_2$$

Christoffel load

$$\Gamma_{\gamma(s)}(\Delta, \Delta)$$

$$+ \nabla_w Q_F$$

$$+ \text{CG on } F + \lambda I$$

Operator norms

$$\|\partial_z F\|_{\text{op}}$$

power iteration

$$\xi_G = \mathcal{G}h \text{ (Green kernel, } O(M) \text{ not } O(M^2)) \Rightarrow D_{\text{rel}} = \sup_t \|\xi_G(t)\|_2 / \|\Delta\|_2$$

Not interchangeable: Hutchinson traces answer a typical-direction question, power iteration a worst-case one. A random probe returns

$$\mathbb{E} z^\top A z = \text{tr} A / P = \|A\|_{\text{op}} r_{\text{eff}}(A) / P, \text{ so with } NK \ll P \text{ it reports a flattening exponent that is too large.}$$